



System Design

Given a customer scenario, define an environment (sandbox) strategy that utilizes the correct sandbox types. (e.g. multiple project streams, training requirements, staging, production, and hotfixes).

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After studying this topic, you should be able to:

-  Describe the main features of Salesforce sandboxes.
-  Describe the features of the four types of sandboxes.
-  Identify the refresh intervals and storage limits available for the four types of sandboxes.
-  Define an appropriate environment strategy that utilizes the correct sandbox types for the given customer requirements.



Introduction

A **Salesforce sandbox** is a separate, isolated environment that contains some or all of the data and metadata from production. Sandboxes can be used for **development, testing, staging, and training**. There are four types of sandboxes, namely: **Developer, Developer Pro, Partial Copy, and Full**. Each type of sandbox is used for specific use cases.

A new sandbox is created with the **metadata and data (depending on the sandbox type)** from the production org. **Refreshing a sandbox** updates its metadata from the resource org. If the sandbox is a **clone** or uses a **sandbox template**, the process updates the org's data and its metadata. Sandboxes have different **refresh intervals** and **storage limits**.



Sandbox Strategy

Developer Sandbox

A Developer sandbox can be used for development and testing in an isolated environment.

Developer Pro Sandbox

A Developer Pro sandbox can be used for quality assurance, integration testing, and user training.



Partial Copy Sandbox

A Partial Copy sandbox can be used for user-acceptance testing, integration testing, and testing.

Full Sandbox

A Full sandbox can be used for performance testing, load testing, and staging.

Salesforce Sandboxes

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Salesforce Sandboxes

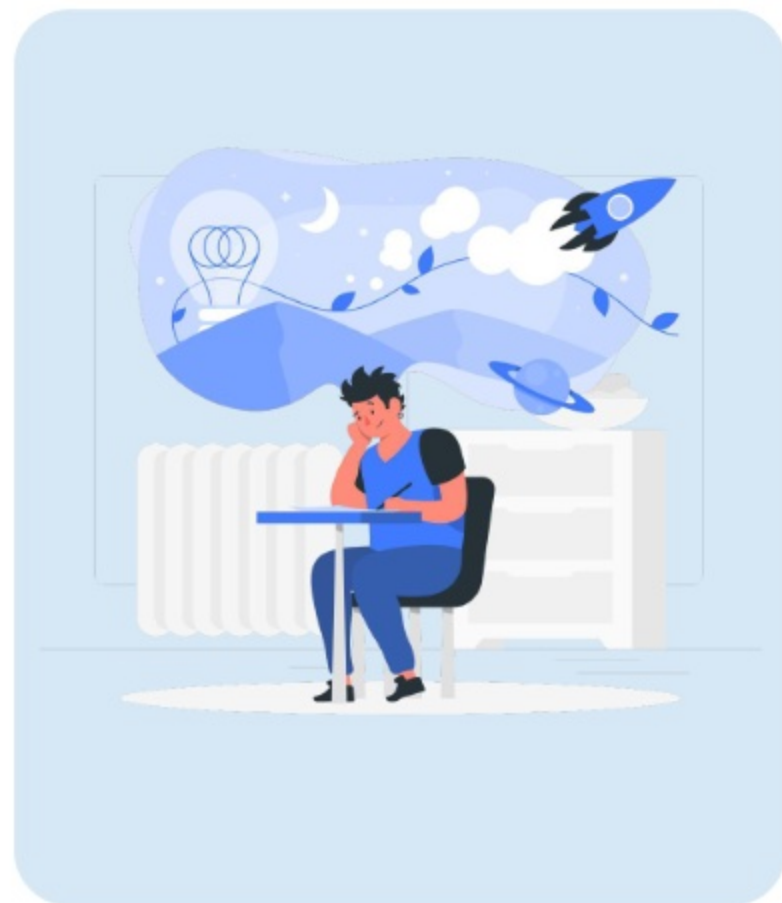
A **Salesforce sandbox** is a separate, isolated environment that contains **some or all of the data and metadata** from production.

❖ USAGE

Sandboxes can be used for **development, testing, staging**, and **training**, without affecting the data and metadata in the production org.

❖ DEVELOPMENT & TESTING

Sandboxes allow **isolated customization and development** work until changes are ready to be deployed. Changes can also be **tested** against copies of the production data and users.



Salesforce Sandboxes

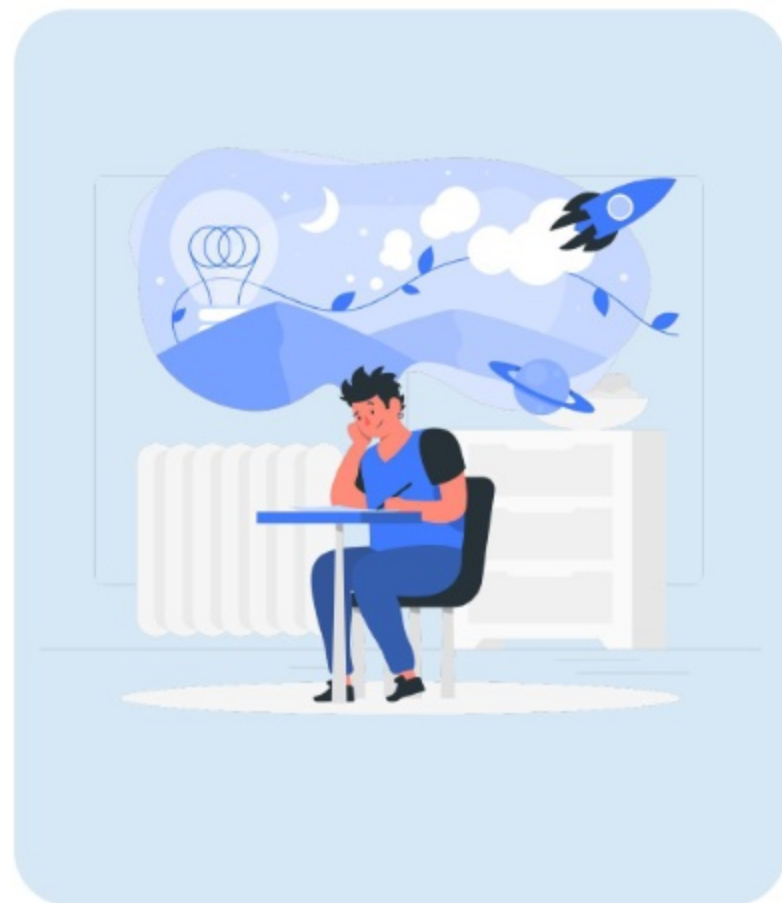
A **Salesforce sandbox** is a separate, isolated environment that contains **some or all of the data and metadata** from production.

❖ TRAINING

A sandbox can be used as a **training environment** for end users before an application is deployed to production.

❖ DEPLOYMENT

Individual changes from administrators and developers can be **coordinated and deployed** to production.



Sandbox Types

There are **four types** of Sandboxes.



Developer

A Developer sandbox includes a copy of the production org's **metadata**. It is typically used for isolated **development and testing**.



Developer Pro

A Developer Pro sandbox includes a copy of the production org's **metadata** and can host **larger datasets**. It is typically used for more **development and quality assurance** tasks, **integration testing**, and **user training**.



Partial Copy

A Partial Copy sandbox includes a copy of the production org's **metadata** and a **sample** of **production data** defined by a **sandbox template**. It is typically used for **user acceptance testing**, **integration testing**, and **training**.



Full

A Full sandbox is a **replica** of the production org and includes **all data and metadata**. It is used for **performance testing**, **load testing**, and **staging**.

Creating Sandboxes

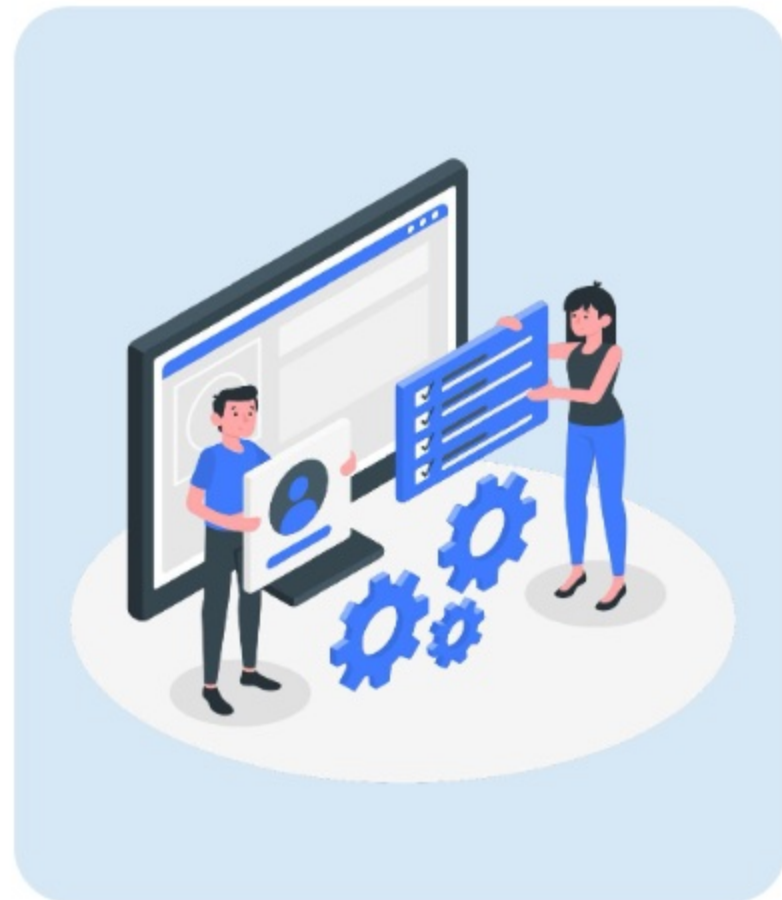
Creating a sandbox **copies the metadata and data (depending on the sandbox type)** from the production org to the new sandbox org.

❖ NEW SANDBOX

A new sandbox can be created from the **Sandboxes** page in Setup. It requires choosing a **name, description, and sandbox type**.

❖ ADDITIONAL SANDBOXES

If a **sandbox option is not visible** or **more licenses are needed**, Salesforce should be contacted to order **additional sandboxes**.



Creating Sandboxes

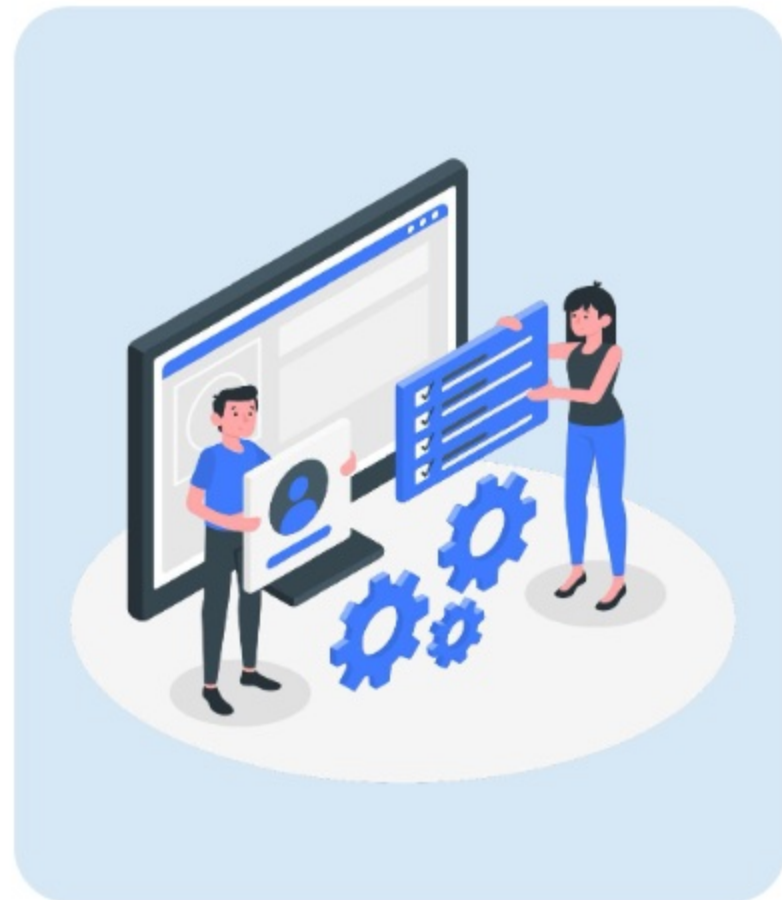
Creating a sandbox **copies the metadata and data (depending on the sandbox type)** from the production org to the new sandbox org.

❖ PURCHASED SANDBOXES

If the number of **purchased sandboxes is reduced**, the number of sandboxes should be **matched** to the purchased number. For example, if there are two Full sandboxes but only one was purchased, a Full sandbox cannot be created.

❖ SANDBOX TEMPLATE

For **Partial Copy** and **Full** sandboxes, a **template** can be selected for the data to include. For a Full sandbox, **all data** can be included; the amount of **field history tracking data** to be included can be selected, and **whether to copy Chatter data** can be selected.



Creating Sandboxes

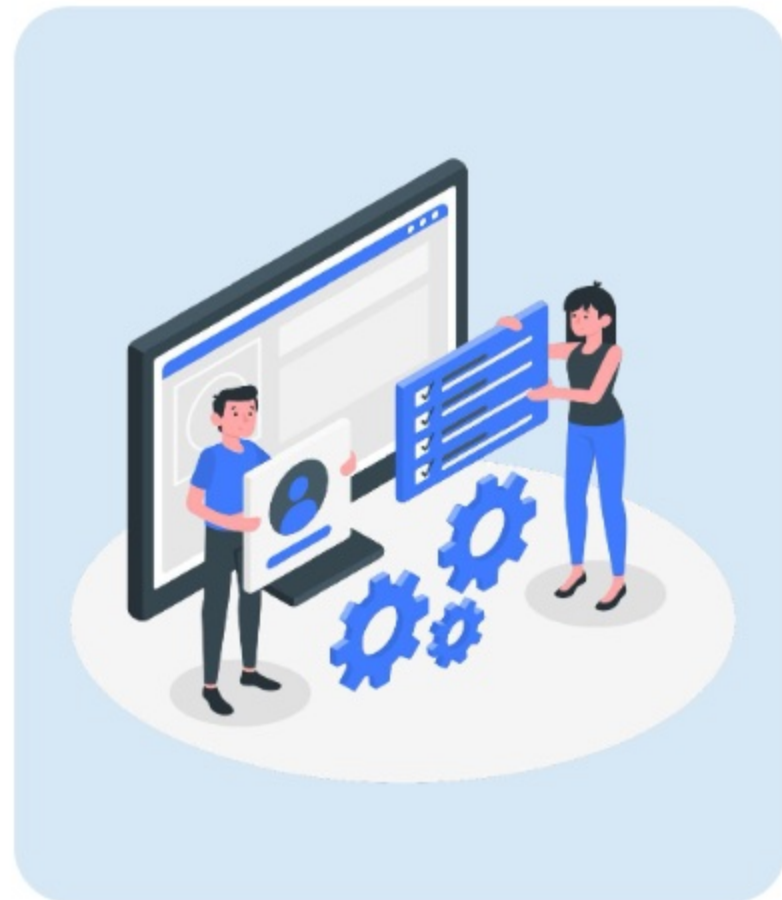
Creating a sandbox **copies the metadata and data (depending on the sandbox type)** from the production org to the new sandbox org.

❖ ADDITIONAL OPTIONS

To **run scripts** after each create and refresh for the created sandbox, an **Apex class** created for the **SandboxPostCopy** interface can be specified. A **public user group** that contains the users that require access to the sandbox can be selected.

❖ PROCESSING TIME

The process of creating a sandbox takes from **several minutes to several days**, depending on the **size and type of the sandbox**. When the sandbox is ready, a **notification email** is sent.



Refreshing Sandboxes

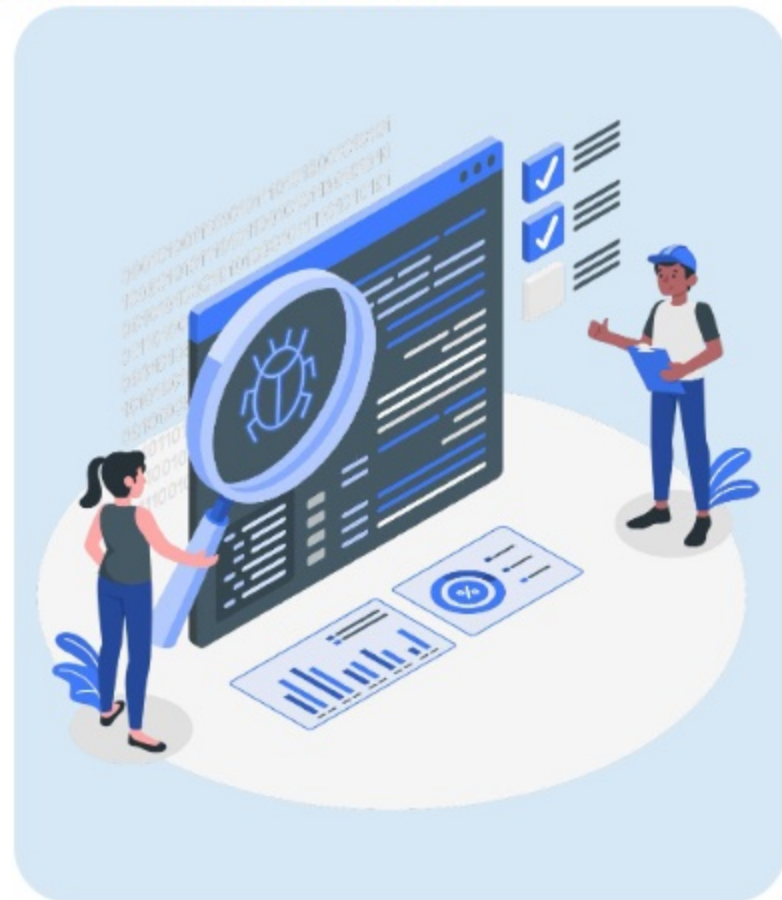
A sandbox can be **refreshed** to **update its metadata** from the resource org. If the sandbox is a **clone or uses a sandbox template**, the process updates the org's **data and its metadata**.

❖ REFRESHING A SANDBOX

Sandboxes that can be refreshed are visible on the **Sandboxes** page in Setup. A **Refresh** link appears next to the next of a sandbox that can be refreshed.

❖ OPTIONS

Name, Description, and Create From values can be reviewed and edited. If the sandbox is **not a clone**, the **type of sandbox environment** required can be selected.



Refreshing Sandboxes

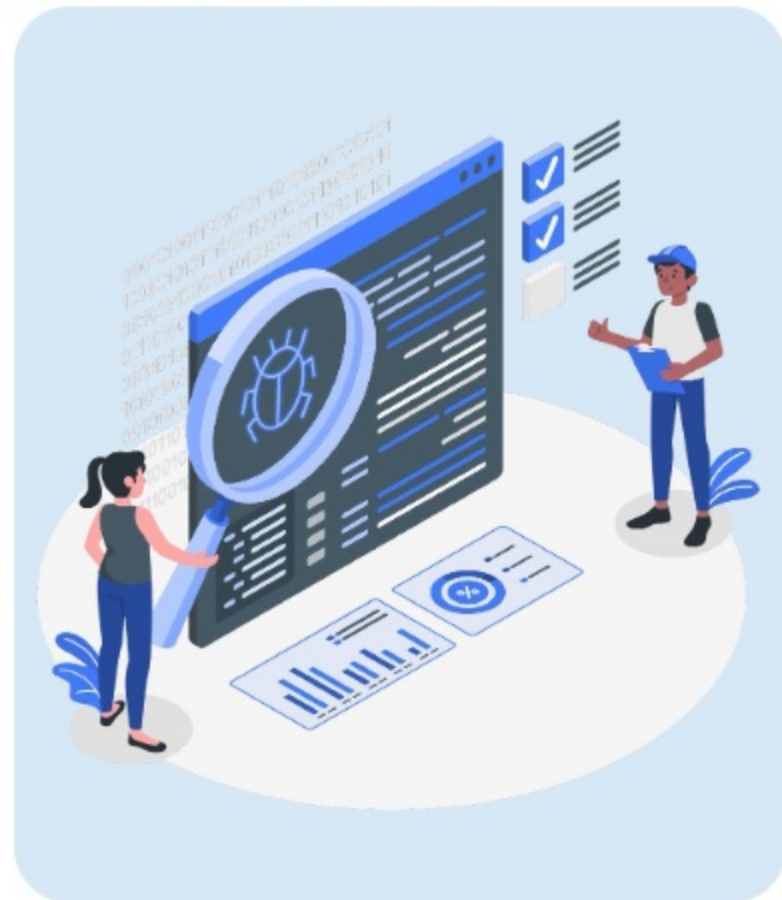
A sandbox can be **refreshed** to **update its metadata** from the resource org. If the sandbox is a **clone or uses a sandbox template**, the process updates the org's **data and its metadata**.

❖ DATA

The data that should be **copied** can be selected. For **partial copy and full sandboxes**, **template-based data** can be selected. For a **Full** sandbox, the amount of **field history tracking data** to be included can be selected, and **whether to copy Chatter data** can be selected.

❖ AUTO-ACTIVATION

The sandbox can be auto-refreshed after refreshing it by selecting **Auto Activate**. An **activation email** is not sent if this options is selected.



Refreshing Sandboxes

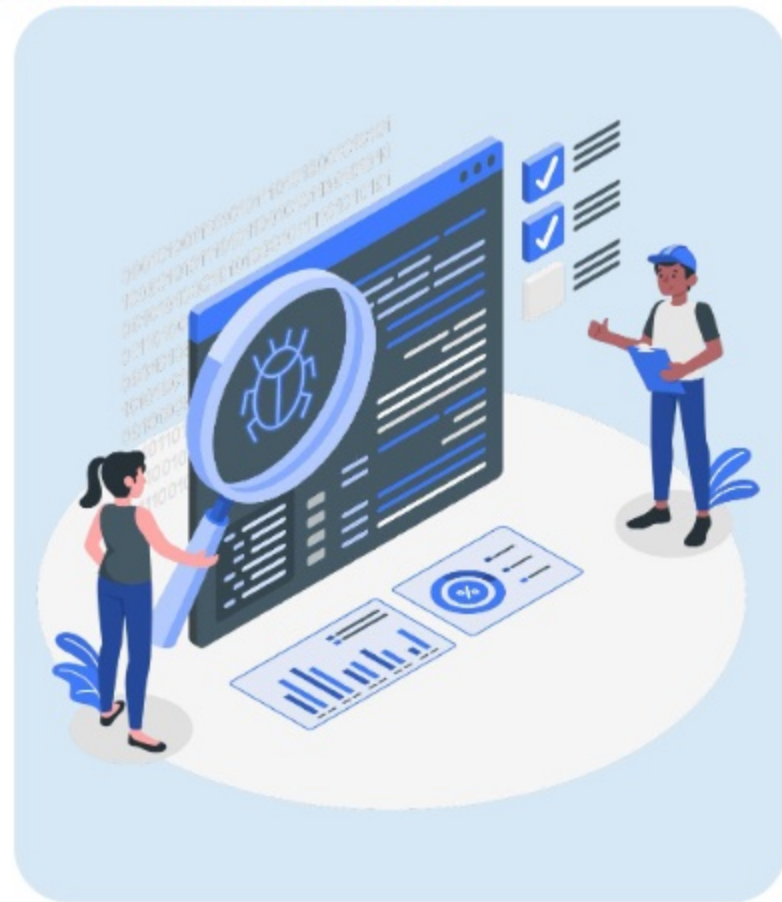
A sandbox can be **refreshed** to **update its metadata** from the resource org. If the sandbox is a **clone or uses a sandbox template**, the process updates the org's **data and its metadata**.

❖ ACTIVATION

A refreshed sandbox must be **activated** before it can be used. The **'Activate'** link next to the name of a sandbox can be selected.

❖ DISCARDING

The refreshed sandbox can also be **discarded**. A discarded sandbox **cannot be recovered**, and discarding a refreshed sandbox **reverts to its previous version, deleting the new version**.



Sandbox Refresh Intervals and Storage Limits

Different sandbox environments can be created based on the needs for **storage, configuration, and frequency of refresh**.

SANDBOX TYPE	REFRESH INTERVAL	STORAGE LIMIT
Developer	1 day	200 MB data storage and 200 MB file storage
Developer Pro	1 day	1 GB data storage and 1 GB file storage
Partial Copy	5 days	5 GB data storage and file storage same as production
Full	29 days	Data storage and file storage same as production

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Healthcare, a leader in health technology solutions, is enhancing its Salesforce platform to support expansive growth and the integration of advanced health applications. It currently manages its operations using Sales Cloud, Service Cloud, and Experience Cloud. Its development teams are currently working on multiple projects. It is in the process of introducing new applications and services, making the need for the optimized use of Salesforce sandboxes critical.

The Salesforce architect of the company is tasked with crafting environment strategies for the company's requirements by utilizing the correct sandbox types for aspects such as multiple project streams, training, staging, and hotfixes.

Requirements & Solutions

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Requirement 1 (Part 1)



SCENARIO

The company needs to manage multiple concurrent development projects for Sales Cloud and Service Cloud. The development teams face challenges in environment management due to upcoming major releases and differing release timelines for Sales and Service Cloud enhancements. The Sales Cloud development team uses a Developer Pro sandbox for development and testing. The Service Cloud development team needs to use a part of the Sales Cloud work from the sandbox. The company requires an effective sandbox strategy to accommodate these development efforts without compromising the quality or timeline of either project.



Requirement 1 (Part 2)



SCENARIO

In addition, to ensure that end users are properly trained to use the new features and enhancements, the company would like to conduct weekly training sessions. Each training session will last for five days and train a new set of trainees on suitable samples of production data. All the training data should be deleted at the end of the week. The architect must suggest an optimal sandbox strategy for the training sessions.



Solution 1



SOLUTION

A new **Developer Pro** sandbox can be created for the Service Cloud development team by cloning the existing sandbox currently used by the Sales Cloud development team. This approach would allow both teams to work independently on their requirements without interference.

For the training sessions, a **Partial Copy sandbox** can be used. It can be **refreshed every weekend**. An appropriate **sandbox template** can be selected to include a sample of training data from production.



Requirement 2



SCENARIO

The company would like to perform thorough batch data testing and a final review of new features and enhancements before deployment to production. The sandbox strategy for batch testing should consider the production data required. Furthermore, there are currently two Full sandboxes. One is used for user acceptance testing, and the other was used for performance testing in the past but is no longer used. A Salesforce administrator has reported that none of the sandboxes can be refreshed. The architect needs to recommend a suitable action to resolve the issue.



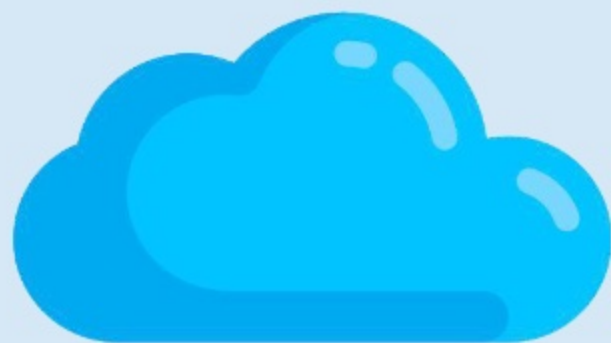
Solution 2



SOLUTION

A **Full sandbox** can be used for staging. The same can be used for **batch data testing** if the testing scenarios require all the production data. A **Partial Copy sandbox** can be utilized for testing scenarios that require only a subset of production data.

The current sandbox utilization should be reviewed to resolve the issue with sandbox refreshes. The **Full sandbox** that was previously used for performance testing and is no longer being used should be **deleted** to align the organization's sandbox usage with its license entitlements and restore the ability to refresh sandboxes as needed.



Requirement 3



SCENARIO

A development team that works on the company's Experience Cloud site has identified the need for a hotfix environment to resolve urgent production issues. In addition, the team regularly needs to test its integrations between the site and multiple enterprise systems, including an HRIS. The sandbox required for testing integration performance should contain up to 10,000 records of each object. The architect needs to suggest the sandbox types and workflow that should be used for these two use cases.



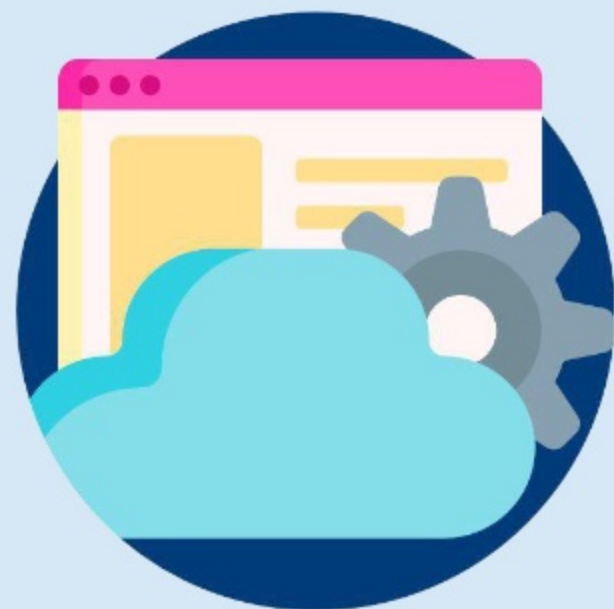
Solution 3



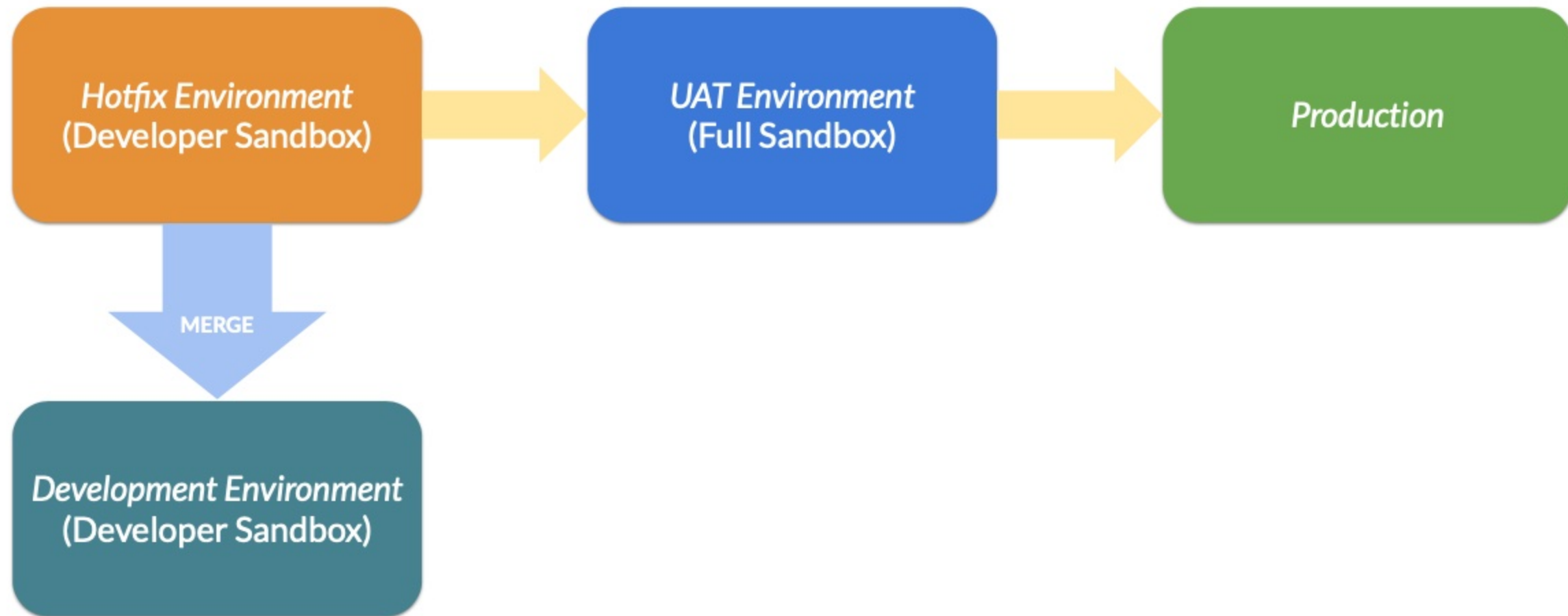
SOLUTION

A **Developer sandbox** can be set up to work on urgent bug fixes. Once development and unit testing have been completed in this sandbox, changes can be pushed to a **Full sandbox** for **user-acceptance testing**, and then deployed to production. Once a fix has been deployed to production, it can be **merged** with the downstream environments to maintain a complete codebase.

For the purpose of testing integrations with enterprise systems, a **Partial Copy sandbox** can be created using a sandbox template with all the objects required for testing.



Solution 3

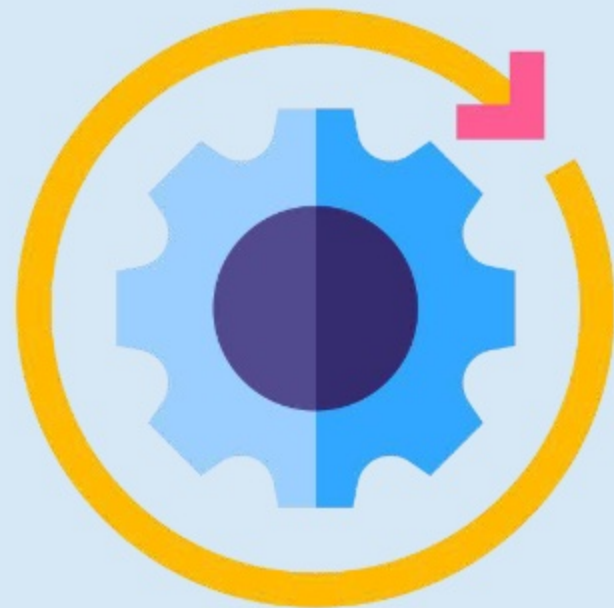


Requirement 4



SCENARIO

The current development lifecycle used by the company starts with Developer sandboxes and moves to a Developer Pro sandbox for integration testing, a Partial Copy sandbox for QA testing, a Full sandbox for UAT, and then production. The company's CTO would ask the architect to recommend a sandbox refresh plan that the development teams can use as a reference. It should describe how often each type of sandbox should be refreshed.

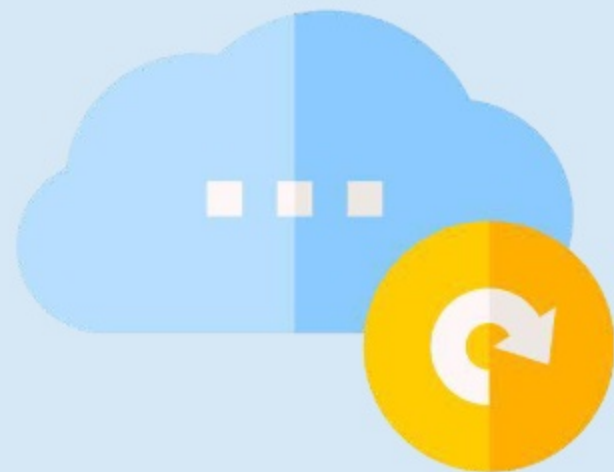


Solution 4



SOLUTION

A sandbox should be refreshed when it is necessary to **update its metadata from the source org**. For example, a Developer sandbox can be refreshed to replicate several configuration updates made to the production org as a result of a separate development stream. If a sandbox is a **clone** or uses a **sandbox template**, the refresh process updates the data and metadata of the sandbox. **Developer** and **Developer Pro sandboxes** can be refreshed **every day**. A **Partial Copy** sandbox can be refreshed **once every 5 days**. A **Full** sandbox can be refreshed **once every 29 days**.



Learn More



[When to Use a Sandbox](#)



[Hotfix](#)

